

two-wire design but upgrades it for today's needs. It's backward compatible with I2C, which means older devices can still connect, but it offers major enhancements like faster communication speeds, Dynamic (runtime) Address Assignment, In-Band Interrupts, and lower power operation.

For server platforms, this means better performance for the BMC and faster sensor polling — all with fewer wires and less power draw. I3C is becoming an integral part of next-generation server management and data centre monitoring infrastructures.

I3C bus uses both open-drain and push-pull. Open-drain is used for initialisation and arbitration, while data transfer is used for push-pull.

In push-pull a device can actively drive the line both LOW and HIGH. It connects the line to the ground to send a logic 0 and connects it to the supply voltage to send a logic 1.

Push-pull outputs do not require pull-up resistors and are typically used in point-to-point or single-driver communication interfaces.

Open-drain to push-pull switching in I3C is as follows:

- The I3C bus always starts in open-drain mode to maintain backward compatibility with legacy I2C devices and to support arbitration during bus initialisation.
- Once dynamic addresses are assigned and only one target is selected for a private transfer, arbitration is no longer required.
- In this phase, the master and the selected I3C target may use push-pull signalling on the SDA line to achieve higher data rates and faster signal transitions.
- If legacy I2C devices are present on the bus, they do not respond to CCC (common command codes) such as ENTDAAs. The master must continue to use open-drain signalling for bus management and any I2C-compatible transactions.

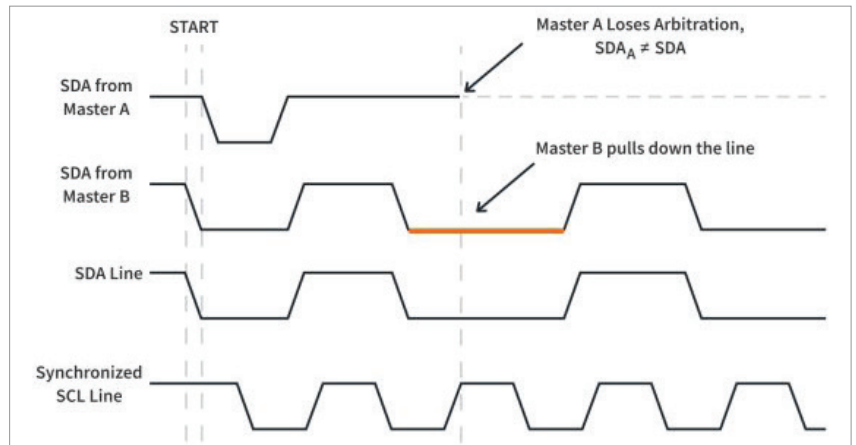


Figure 5: I2C arbitration

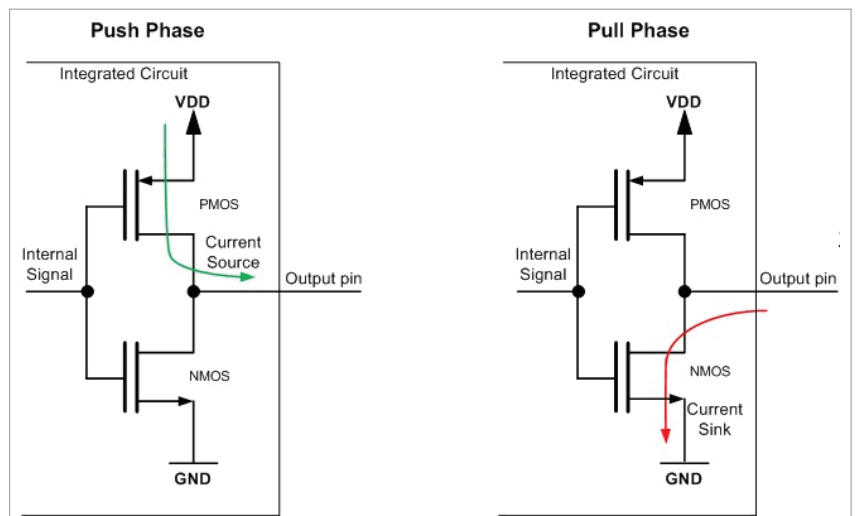


Figure 6: Push-pull

- However, push-pull signalling can still be used for private I3C transfers to I3C targets, while ensuring that electrical compatibility with I2C devices is maintained.

During ENTDAAs (Enter Dynamic Address Assignment), devices do not yet have addresses. So for identifying the devices, they use PID (48 bits). PID contains:

- Vendor ID (who made the chip)
- Part ID (model of the chip)
- Instance ID (which copy of the chip it is)

All chips have a globally unique identity. This is how arbitration works safely.

Dynamic Address Assignment flow

- **Master broadcasts ENTDAAs:** All devices without a dynamic address prepare to identify themselves.
- **All unaddressed targets prepare to respond:** Each target loads its 48-bit PID into shift registers. The device copies its internal 48-bit ID into hardware that can send one bit per clock on the SDA line and monitor the bus for arbitration, which is required for DAA in I3C.
- **PID arbitration:** The device with the lowest PID value wins. The master receives the full PID of the winning device.